

LightCity

Analysis Document



...**LIGHTCITY**...

Problem Analysis

The current common light systems are not communicating with each other and it takes time to notice a broken light and fix it. Current street lights also operates at either on or off state, and during the on state it operates at the set light intensity without adjusting to the environment so it uses excess energy to light up empty pavements.

Project Goal

The project is aimed to create a decentralized lighting system that continuously communicates with each other to improve energy efficiency while maintaining effective lighting for the users. To maximize the efficiency of maintenance like noticing immediately whenever a bulb is broken, a monitor will be able see if a light is malfunctioning and act accordingly. The neighboring lights should be able to compensate for a faulty node.

Must Requirements

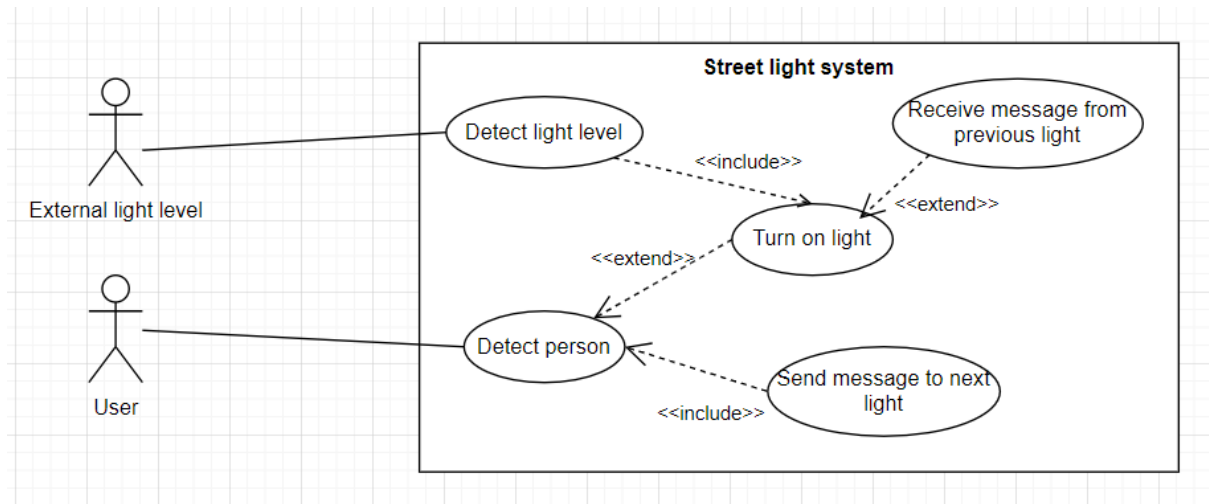
1. Neighboring nodes compensate for a faulty node.
2. The light whose sensor is triggered, should turn on at full intensity and notify the neighboring lights.
3. The lights moving away from the sensor that was triggered, should progressively dimmer until reaching minimum brightness.
4. When the light outside reaches a certain maximum brightness threshold, the lights should be turned off.
5. When the light outside reaches a certain minimum brightness threshold, the light should be turned on at the minimum brightness.

Use Cases

Adjust brightness	
Description:	Whenever a person is detected the brightness of the lights will automatically adjust according to the position of the person.
Actor:	User

Adjust lights according to environment	
Description:	When light levels go down(sun is setting), turn on lights which detect low light level to a level that provides minimal vision.
Actor:	External light level

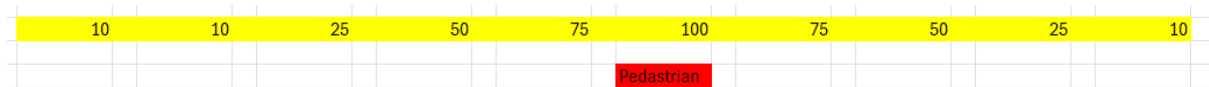
Turn on lights during network outage	
Description:	Lights will be on at 75% intensity or off depending on light level to ensure lighting is provided when communication has failed.
Actor:	External light level



Scenarios:

Happy Flow 1:

A single pedestrian is walking on the pavement underneath the LightCity Streetlights. Once the streetlight detects movement it starts shining at 100% intensity, the next light at 75%, the one after that at 50%, then 25% and the one after that at 10%. Behind him it is the same.



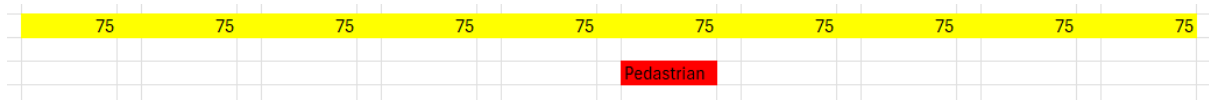
Happy Flow 2:

When two pedestrians approach each other from opposite directions, the streetlight directly above each pedestrian shines at 100% brightness, and the surrounding lights follow the pattern (next light at 75%, the one after that at 50%, and so on). As they get closer and their lighting zones overlap, the streetlights prioritize maintaining clear visibility. For any streetlight between them, the brightness values combine, ensuring optimal illumination where their lighting zones meet. For example, if pedestrian 1 is under light ONE at 100% and pedestrian 2 is under light FOUR at 100%, the lights between them (TWO and THREE) will balance to 75%, ensuring smooth lighting for both pedestrians. After they pass each other, the streetlights revert to the normal pattern, with each pedestrian's path lighting up independently.



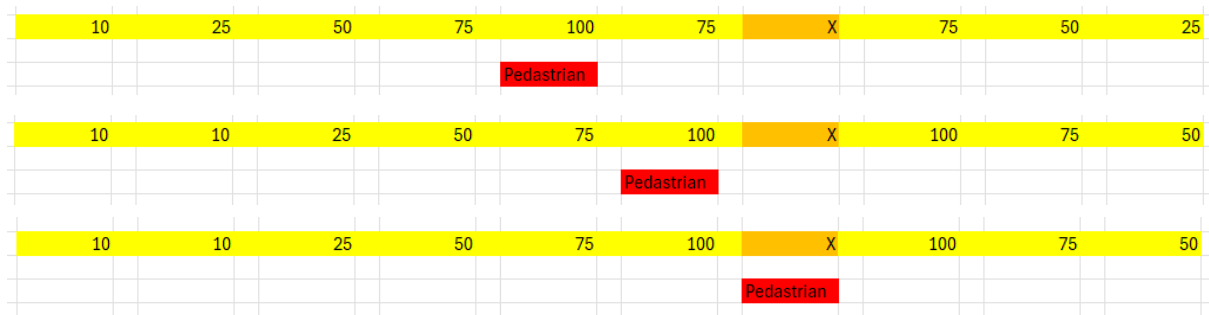
Unhappy Flow 1:

When the entire network crashes, the lights should all go with an intensity of 75%, unrelated to the time or day. This ensures that even when it is at night, there is still enough visibility to maintain a safe environment.



Unhappy Flow 2:

When a light breaks down or a connection with a node fails, the lights intensity should shift to the next nodes/streetlights. An example would be 10%, 25%, 50%, 75%, 100%, 75%, broken, 75%, 50%, 25%, 10%, etc.



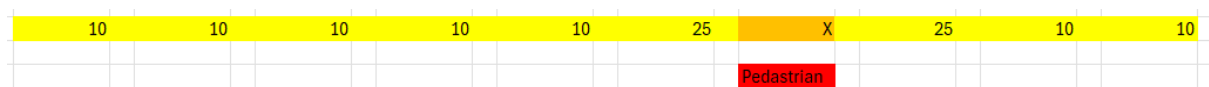
Bordercase Scenario 1:

A single pedestrian walks out of its house and starts walking on the pavement underneath the LightCity Streetlights, but both streetlights detect the same person. They both communicate with the next light and be at 100% until the person exits the range of the sensor and it act as a normal flow.



Bordercase Scenario 2:

When a light breaks down or a connection with a node fails, the lights intensity should shift to the next nodes/streetlights. An example would be 10%, 25%, 50%, 75%, 100%, 75%, broken, 75%, 50%, 25%, 10%, etc. But when the person is underneath the broken light before the sensor has detected any movement, the lights will be as follows:



This will stay until the next light detects movements and every streetlight will act as normal. This may require some more thought.

